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MILITARY SPECIFICATION

CABINETS, FOOD SERVICE, WARMER-REFRIGERATOR, ROLL-THRU, ROLL-IN

This limited coordination military specification has been prepared by the US Army Natick Research and Development Command, Natick, MA 01760, based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-C-43427A. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers cabinets roll-thru and roll-in, warmer and refrigerator and combinations thereof.

1.2 Classification. Cabinets covered by this specification shall be of the following types, styles, classes, duties and models as specified (see 6.2).

Type I - Refrigerator cabinet.

Class 1 - Single cabinet.
Class 2 - Double cabinet.
Class 3 - Triple cabinet.
Class 4 - Four section cabinet.

Inclosure 1 - Self-contained condensing unit.
Inclosure 2 - Remote condensing unit.

Duty A - Water cooled condensing unit.
Duty B - Air cooled condensing unit.

Type II - Warmer cabinet.

Class 1 - Single cabinet.
Class 2 - Double cabinet.
Class 3 - Triple cabinet.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRXNM-TDG, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

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Type III - Combination cabinets.

Class 1 - 1 warmer - 1 refrigerator.
Class 2 - 1 warmer - 2 refrigerator.
Class 3 - 2 warmer - 1 refrigerator.
Class 4 - 2 warmer - 2 refrigerator.

Inclosure 1 - Self-contained condensing unit.
Inclosure 2 - Remote condensing unit.

Duty A - Water cooled condensing unit.
Duty B - Air cooled condensing unit.

Style A - Non-insulated, corrosion-resistant bottom with wall type cart guides.
Style B - Two-inch deep insulated, corrosion-resistant bottom with wheel groove cart guides.
Style C - Three-inch deep insulated bottom with wall type cart guides for installation in a recessed floor 3-3/8 inch maximum depth.
Style D - Bottomless cabinet with wall cart guides.

Model A - Roll-in.
Model B - Roll-thru.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-F-1421 - Fluorocarbon Refrigerants.
HH-I-524 - Insulation Board, Thermal (Polystyrene).
TT-P-664 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-R-43399 - Rack, Mobile (For Roll-Thru Refrigerators and Roll-Thru Food Warmers).
MIL-R-12323 - Refrigerators and Related Equipment, Packaging and Packing of.

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STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 36 - Structural Steel
A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
B 380 - Seamless Copper Tube for Air Conditioning and Refrigeration Field Service

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 197 - Commercial Electric Cooking Appliances.
UL 471 - Commercial Refrigerators.
UL 303 - Condensing Units Refrigeration.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I., New York 11746; 1655 Scott Boulevard, Santa Clara, CA 95050, or 2602 Tampa East Blvd., Tampa, FL 33169.)

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NATIONAL SANITATION FOUNDATION (NSF)

Standard 4 - Commercial Cooking and Hot Food Storage Equipment.
Standard 7 - Food Service Refrigerators and Storage Freezers.
Listing of Food Service Equipment.

(Application for copies should be addressed to the National Sanitiation Foundation, Ann Arbor, MI 48106.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standard MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

NATIONAL BUREAU OF STANDARDS (NBS)

Standard for Devices to Permit the Opening of Household Refrigerator Doors from the Inside.

(Application for copies should be addressed to the Director, National Bureau of Standards, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The cabinets shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The cabinets shall conform to the applicable requirements of NSF Standards 4 and 7, and UL 197, 303 and 471.

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3.2.1 Compliance. Prior to commencing production the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the cabinets he proposes to furnish under this specification meet the applicable requirements of NSF Standards 4 and 7, and UL 197, 303 and 471.

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished cabinet, or
(2) A certification for the cabinet issued by NSF under their special one-time contract evaluation/certification service, or
(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the cabinets are being procured, indicating that the cabinets have been tested and conform to the applicable requirements of NSF Standards 4 and 7.

3.2.1.2 UL. Acceptable evidence of meeting the requirements of UL shall be the UL listing mark or label, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the cabinets offered have been tested and conform to UL 197, 303 and 471.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial dispensers provided the completed item complies with all provisions of this specification.

3.3.1 Stainless steel. Stainless steel shall conform to type 302 and 304 of ASTM A 167.

3.3.2 Structural steel. Structural steel shall conform to ASTM A 36.

3.3.3 Refrigeration tube. Tube used for refrigerant lines shall conform to ASTM B 280. Tube used for the fabrication of the evaporator and condenser shall be the type normally used for the purpose by the manufacturer except it shall have a wall thickness not less than 0.016 inch (.40 mm) for 3/8 inch (.95 mm) and 1/2 inch (12.7 mm) O.D. tube.

3.3.4 Insulation. All insulation shall be expanded polyurethane or polystyrene, or industrial type fiberglas having a "K" factor determined by using a 75°F (238°C) mean temperature, expressed in British thermal units (Btu) per hour per degree F temperature difference per inch of thickness. Loose fill type insulation shall not be acceptable.

3.3.4.1 Pre-expanded insulation. Pre-expanded insulation shall be polyurethane material conforming to class 2, types I or II of HH-I-524.

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3.3.4.2 Foamed-in-place insulation. Foamed-in-place insulation shall be polyurethane material having a manufacturer's commercial rated "K" factor of not more than 0.015, a core density of not less than 1.9 pounds (.86 kg) per cubic foot (.028 m^3) when tested as specified in 4.4.2.

3.3.4.3 Semirigid insulation. Semirigid insulation shall be "B" fiber or finer industrial type fiberglas with a manufacturer's commercial rated "K" factor of not more than 0.260 and a density of at least 1.0 (.45 kg) but not more than 3.0 pounds (1.36 kg) per cubic foot (.028 m^3) and shall be held together with a thermosetting binder.

3.3.5 Refrigerant. The refrigerant shall be type 12 or type 22 conforming to BB-F-1421.

3.4 Design. The cabinets shall be designed as integral free standing units in the types, classes, inclosures, duties, styles, and models specified in 1.2. All cabinets shall be designed with interior lighting, thermometers, switches, heaters, condensing units, and controls necessary for automatic operation. The refrigerator cabinets or sections shall automatically maintain temperatures at 38 degrees Fahrenheit (°F) (-17.7 °C) plus or minus 2°F (1°C) when tested as specified in 4.4.1. The warmer cabinets or sections shall automatically maintain temperatures at 150°F (65.5°C) to 160°F (71.1°C), plus or minus 2°F (1°C) when tested as specified in 4.4.1. The self-contained refrigerator cabinets or sections shall house the condensing unit in the top of the cabinet. The cabinet dimensions shall be as shown on figure 1. The door opening on each section shall be 65 to 66 inches (1.65 to 1.67 m) by 26 to 27 inches (.66 to .69 m). Each cabinet or section shall accomodate mobile racks conforming to MIL-R-43399. Unless otherwise specified (see 6.2), a rack shall be furnished with each section or cabinet. All major assemblies and installed attachments shall be accessible for maintenance repair or replacement. Removable access covers and plates shall be provided as necessary to permit adjustment, service and replacement can be accomplished without disturbing or disassembling any major component or equipment. When specified bumpers shall be furnished on model A and model B cabinets (see 3.5.8).

3.5 Construction. The cabinets shall be constructed as specified herein and as illustrated in figure 1. Figure 1, except for dimensions (width dimensions are multiples of the width of a single cabinet), is for illustrative purposes only and is not intended to preclude other cabinets otherwise meeting the requirements of this specification. The style A cabinets shall be constructed having a non-insulated, corrosion-resistant bottom with wall cart guides for installation on a floor surface. The style B cabinets shall be constructed having a 2 inch (50.8 mm) deep insulated corrosion-resistant bottom

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with wheel groove cart guides for installation on a floor surface. The style C cabinets shall be constructed having a 3 inch (76.2 mm) deep insulated bottom with wall cart guides for installation in a recessed floor or pit a maximum depth of 3-3/8 inches (85.7 mm). The style D cabinets shall be constructed having no bottom with wall cart guides for installation on a floor surface. The inner liner and outer shell of the cabinets shall be fabricated of stainless steel specified in 3.3.1 having a thickness of 0.0359 inch (.91 mm) or greater. The space between the inner liner and outer shell shall be completely filled with the insulation specified in 3.3.4.3 when framed construction is used; when frameless construction is used the insulation specified in 3.3.4.1 or 3.3.4.2 shall be used. The interior floor for styles A, B, and C shall be constructed of steel specified in 3.3.2, having a thickness of 0.0598 inch (1.52 mm) or greater. Framing members, when used, shall be constructed of steel specified in 3.3.1, 3.3.2, or galvanized steel. When steel is used it shall be phosphatized and coated with a paint primer conforming to TT-P-664. Each section of the cabinets shall have full length doors with hardware. All doors shall be insulated with insulation specified in 3.3.4.1 or 3.4.2 and gasketed. Openings in the inner liner and outer shell for refrigerant tube, piping, and electrical fittings shall be mechanically sealed upon completion of the installation to prevent moisture infiltration into the insulation. The location of the openings in the cabinets shall conform to the manufacturers standard practice unless otherwise specified (see 6.2). Interior rack guides, constructed of stainless steel specified in 3.3.1 having a thickness of 0.0598 inch (1.52 mm) or greater shall be furnished in each section.

3.5.1 Electrical characteristics. Unless otherwise specified (see 6.2), the cabinets shall operate on a nominal 208-volt, 1-phase, 60 hertz power source. Electric motors shall conform to NEMA MG-1. All accessories may be operated on a nominal 120-volt, 1-phase, 60 Hertz power source. When 115-volt, 1-phase, 60 hertz power source is specified, the cabinets shall be equipped with a three conductor cord, 6 foot (1.8 m) minimum length and three prong grounding plug. The cabinets shall be furnished completely wired and ready for connection to electric power source.

3.5.2 Protection against electrolytic or galvanic corrosion. Joints between dissimilar metals which may become wet in service shall be protected against electrolytic or galvanic corrosion by proper selection of materials, by coating, by isolating, by gasketing, or by similar suitable means. Bolts, nuts, pins, screws and other fastenings and fittings of metal shall be of the same material as the materials being joined or shall be cathodic to the materials being joined.

3.5.3 Breaker strips. For refrigerator cabinets or sections breaker strips of thermoplastic material, thermosetting plastic, natural rubber, or synthetic rubber shall be installed to join the outer shell and inner liner around exterior door openings. Joints in breaker strips shall be smooth and cleanable; edges of the breaker strips shall be eased or rounded; and the entire installation shall provide a smoothly finished appearance.

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3.5.4 Condensate disposal. A condensate evaporator pan and a drain pipe shall be provided in the refrigerator sections. The condensate evaporator pan shall be stainless steel, aluminum, or glass. The drain pipe shall be brass or copper tubing or aluminum and shall include an "S" trap large enough for cleaning or be readily removable, for cleaning. The condensate shall be disposed through the drain pipe to a waste drain outside the cabinet. When specified (see 6.2), condensate evaporators shall be used in lieu of the drain pipe and "S" trap.

3.5.5 Doors. The doors shall be constructed of stainless steel specified in 3.3.1. The door construction shall be frame type or frameless with low heat conductive spacers or separators, designed to prevent sagging or distortion and to provide for mounting of hardware. Gaskets shall provide an even elastic seal around the door and the method of attachment shall provide for easy replacement. Doors shall comply with NBS Standard for "Devices to Permit the Opening of Household Refrigerator Doors from Inside". When specified (see 6.2), door handles shall be recessed. When specified (see 6.2), self-closing doors shall be furnished.

3.5.5.1 Door latch and striker assembly. The door latch and strike assembly, when used, shall be fabricated of steel or malleable iron, copper-nickel-chromium plated; cast or forged brass, nickel-chromium plated; or high pressure die cast zinc, copper-nickel-chromium plated. Nylon contacts may be used in the latch and strike assembly. The door latch and strike assembly with handle shall be adjustable to compensate for reduction in original gasket thickness of at least 10 percent or shall be designed to automatically provide sufficient holding force to effect the gasket seal required over a range of gasket thickness reduction ranging from 0 to 10 percent. Original gasket thickness shall be measured at the latch or handle side of the door with the door closure adjusted by the manufacturer or if necessary, adjusted to effect the gasket seal required.

3.5.5.2 Hinges. Exposed hinges shall be steel or brass, copper-nickel-chromium-plated or stainless steel. Concealed or semi-concealed hinges may be steel finished to match the exterior of the cabinets. Hinges shall be adjustable to compensate for at least a 10 percent reduction of compressible height of original gasket thickness on the hinge side of the door, except where magnetic gaskets are used, adjustment is not required.

3.5.6 Thermometer. A remote reading 2-1/2 inch (63.5 mm) dial type, flush mounted thermometer shall be installed on the front of the cabinets near the top of the door mullions. The thermometers shall have a range of 0°F (-17.7°C) to 60°F (15.5°C) for refrigerated sections, and 100°F (37.7°C) to 200°F (93.3°C) for the warmer sections. The thermometer dial face shall be white with black marking and figures, a colored safety zone and a recalibration feature.

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3.5.7 Thermostat and pilot light. The warmer cabinets or sections shall be equipped with a thermostat calibrated from 100°F (37.7°C) to 200°F (93.3°C), a pilot light, and an "ON-OFF" switch located in the door mullion midway between the door handle a protecting shroud or housing.

3.5.8 Bumpers. When specified (see 6.2), model A cabinets shall be equipped with bumpers on the front and side only and model B cabinets shall be equipped with bumpers on the front and side only. Bumpers shall be fabricated from rubber like material have a durometer hardness of 75 ± 15 on the A scale when tested as specified in 4.2.2.2. Bumpers shall measure 3-1/2 ± 1/2 inches (88.9 ± 12.7 mm) wide by 7/8 ± 3/8 inches (22.2 ± 9.5 mm) deep and shall be firmly assembled to the cabinet at a height of 34 ± 1/8 inches (863 ± 3 mm) from the floor to the centerline of the 3-1/2 inch (88.9 mm) dimension. Bumpers shall be capable of being removed and replaced.

3.6 Refrigeration system. The refrigeration system shall be hermetic type, either thermally or mechanically sealed, or when specified (see 6.2) shall be open type and designed for use with the refrigerant specified in 3.3.5. The refrigeration unit shall be furnished complete with a condensing unit having a receiver, valves, drier, motor, moisture and liquid indicator and all necessary refrigeration and safety controls applicable to the refrigeration system. Provision shall be made for air distribution within the refrigerated space. All components and piping containing refrigerant shall be gas tight (see 4.4.3). The self-contained refrigeration system shall be dried, evacuated, and charged with operating requirements of the refrigerant specified in 3.3.5 and oil recommended by the compressor manufacturer. After charging the refrigeration system shall show dry on the moisture and liquid indicator when tested as specified in 4.4.4. The remote units shall be dehydrated and charged with a holding charge.

3.6.1 Evaporator assembly. The evaporator assembly for the inclosure 1 cabinets shall be either top mounted, located within a top mounted housing, or mullion type, located between the door mullions of the refrigerator section. The evaporator assembly for the inclosure 2 cabinets shall be mullion type, located within the door mullions of the refrigerator section. Each evaporator assembly shall be furnished complete with evaporator fan and fan motor, and all accessories, instruments and controls necessary to provide, and automatically, maintain the refrigerator temperature specified in 3.4. All components of the evaporator assembly shall be contained within a protecting shroud or housing.

3.6.2 Refrigerant controls. Controls shall be provided to automatically maintain the specified cabinet temperatures. Motor cutout switches shall open the motor circuit when the discharge pressure becomes excessively high

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or when the suction pressure becomes excessively low. The automatic control equipment shall be positive in action, and shall be adjustable through the temperature or pressure range required for the temperatures specified in 3.4.

3.6.3 Piping. All refrigerant piping shall be the copper tubing specified in 3.3.3. The tubing shall be arranged so as not to interfere with the servicing or removal of the condensing unit. Fittings shall be of wrought or forged copper. The refrigeration unit design shall prevent condensation on suction lines within the machine compartment. The run of suction line exposed to room air of inclosures 1 and 2 units including the section in the machine compartment of inclosure 2 units shall be insulated with flexible preformed 3/8-inch (9.5 mm) plastic foam material. Linear and butt ends and terminal ends shall be sealed with adhesive recommended for this application by the same manufacturer supplying the insulation. Where liquid and suction lines are fastened together, the insulation shall be installed to cover both lines. All tubing connections shall be soldered, except flared connections shall be provided on expansion valve inlet and outlet connections, dehydrator inlet and outlet connections, receiver outlet valve, and any other locations normal in manufacturer's standard design. Where tubing of different materials are joined, a shrink-fit plastic sleeve or other method shall be used to positively prevent wetting of the surfaces for at least 1-inch (25.4 mm) to each side of the joint.

3.6.4 Electromagnetic compatibility. When specified (see 6.2), the cabinets shall be designed and equipped for electromagnetic compatibility in accordance with class IIB of MIL-STD-461 (see 4.4.5).

3.7 Finish. The cabinets shall have a number 5 finish or better on the exterior fronts, sides, top, doors, and backs; except the roll-in type backs and all exterior floors may be galvanized steel. The interior shall be stainless steel having a number 4 finish or better.

3.8 Identification marking. Identification marking shall be permanently and legibly marked directly on the item or an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved, or applied by photo sensitive means. The plate thickness shall be not less than 0.012 inch (0.3 mm). The identification plate shall include marking requirements of the applicable UL standard.

3.9 Wiring diagram. A wiring diagram shall be glued or otherwise affixed in an accessible location on the cabinet so that it may be read from the front of the cabinet. The diagram shall be waterproof and shall portray each electrical component and its terminals by a sketch, outline, or name, and not by electrical symbol alone, showing the wiring of the cabinets with the wires coded for identification.

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3.10 Workmanship. The finished dispensers shall be complete and clean, and free of fabrication marks, scratches, dents, breaks, sharp edges, sharp corners, deformities, burrs, slivers, or visible evidence of rust.

3.10.1 Rivets. Rivets, when used, shall be tight and the rivet heads and peened ends shall be round and concentric and shall be in full contact with the joined metal surfaces. There shall be no empty rivet holes.

3.10.2 Welds. Welds shall be clean, continuous, sound, free of flux, pits, buildups, undercuts, and shall be ground so as to blend with adjacent surfaces. Spot welds when used shall be sound and free of discoloration, splatter and burn throughs.

3.10.3 Threaded fasteners. Threaded fasteners shall be drawn tight. Lockwashers shall be used on all nuts and bolts. There shall be no crossed or stripped threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except where otherwise indicated herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 In-process examination. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, corrections shall be made to the operation and items affected.

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4.2.2.1.1 Visual examination of operations.

(a) Installation of insulation (see 3.5).
(b) Sealing of openings in cabinet shells against moisture infiltration (see 3.5).
(c) Low carbon steel is primed (see 3.5).
(d) Protection of joined dissimilar metals against corrosion (see 3.5.2).
(e) Internal construction of doors (see 3.5.5).
(f) Installation of piping and accessories (see 3.6.3).

4.2.2.2 In-process testing.

4.2.2.2.1 Density test of insulation. The density test for insulation shall be performed as specified in 4.4.2. The inspection lot shall be the amount of material used in one day's production. The sample unit for this test shall be a 6-inch (152 mm^3) cube of insulation.

4.2.2.2.2 Bumper material hardness. The bumper material shall be tested for hardness in accordance with method 3201 of FED-STD-601 to determine conformance with 3.5.8. The sample unit shall be one bumper. The lot shall consist of all bumpers offered for acceptance at one time. Failure of the sample to comply with the hardness requirement shall be cause for rejection of the lot.

4.2.3 End item inspection. The lot shall consist of all cabinets offered for inspection at one time. The sample unit shall be one completely fabricated cabinet.

4.2.3.1 Visual examination. The cabinet shall be examined for defects in classified in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE I. Classification of visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturers standard product except for changes necessary to comply with specification requirements	X	
Finish	Not as specified	X	
	Rough and sharp edges, burrs, or slivers	X	
Design, construction and workmanship, general	Not self supporting	X	
	Access panels, covers, and plates not provided for maintenance	X	
	Component not as specified	X	
Outer shell and inner liner	Not welded construction	X	
	Openings for tubing, piping and electrical connections not mechanically sealed	X	
Breaker strip	Not installed as specified	X	
	Edges not eased or rounded		X
	Joints not sealed	X	
Doors	Number of doors more or less than specified	X	
	Gasket does not provide even seal	X	
	Gaskets not easily replaced		X
	Not self-closing (when required)	X	
	Handles not recessed (when required)	X	
Evaporator	Not type specified	X	
	Fan or fan motor missing	X	
	Protecting shroud or housing missing	X	
Bumpers	Missing, not type specified	X	
	Not removable		X
Thermometer	Not type or range specified	X	
	Not installed where specified		X

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TABLE I. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Thermostat and pilot lights	Not located as specified		X
Electrical	Cord and plug do not provide grounding (when required)	X	
	Switch location not as specified		X
	Switch not clearly marked	X	
Bolts, nuts, and other threaded fasteners	Broken, stripped, missing, fractured		X
	Not drawn tight		X
Workmanship	Rivets not tight, head and end not round and concentric, not in full contact with joined surface		X
	Sheet metal kinked or bent		X
	Shearing and chipping, ragged not neatly finished		X
	Bolt or rivet holes not accurately drilled or punched		X
	Accessories and piping not securely fastened or braced		X
Welding	Not free from rust scale, paint, grease and other foreign matter	X	
	Scale and flux deposits not removed		X
	Weld not smooth and free of cracks, burnholes, undercuts or incomplete fusion		X
Marking for identification	Not permanently marked on the item or aluminum, brass or stainless steel plate	X	
	Missing, incomplete, illegible	X	
Instructions wiring diagram	Missing, incomplete, illegible	X	

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4.2.3.2 Dimensional examination. Examination shall be made of the cabinets for defects in dimensions. Any dimension not within specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.3.3 End item testing. Each refrigerator cabinet or section, shall be tested in accordance with 4.4.3 and 4.4.4 and failure to pass these tests shall be cause for rejection of the unit. One from the lot shall be tested in accordance with 4.4.1 and 4.4.5, as applicable, and failure to pass these tests shall be cause for rejection of the lot.

4.2.4 Packaging inspection. Examination shall be made to determine that packaging requirements as specified in section 5 of this specification is in conformance with MIL-R-12323.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listed marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Test methods.

4.4.1 Performance test. The performance test shall be conducted in an ambient temperature of 110°F (43°C). The refrigerated sections shall be precooled to a temperature of 38°F (3.3°C), plus or minus 2°F (1°C), while the food warmer sections are maintained at 150°F (65.5°C) to 160°F (71.1°C). The precooling time shall be not less than 5 hours before the test run. The control on the refrigerator cabinets shall be set at 38°F (3.3°C) and the controls shall be set at 155°F (68.3°C) for the warmer cabinets and the test run for a period of 8 hours. During this time observation shall be made that all components function as intended and that the compressor on the refrigerator cabinets does not cycle or cut out on overload protectors. Inability of the cabinets to maintain the temperatures required in 3.2, failure of components to function as intended, or compressor failure shall constitute failure of this test.

4.4.2 Density test for insulation. When foamed-in-place insulation is used, one free foam sample of the insulation specified in 3.3.4.2, a minimum cube of 8-inches (203 mm) on a side, shall be taken prior to each days production. From the center of the sample a cube, 6 inches (152 mm) on a side, shall be cut and weighed. This weight (1/8 of a cubic foot (or 0.00035 m^3)), shall be multiplied by 8 (283 for m^3) to determine conformance with the density requirements of 3.3.4.2. In addition a daily record of the foam insulation shall be kept as follows and made available to the contracting officer.

Calibration of flow (mix proportions)
Amount of flow (per box)
Time of flow (per box)
Mould temperature (per unit)

Any nonconformance with density requirements shall constitute failure of the test.

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4.4.3 Refrigeration leak test. The refrigerator cabinets shall be refrigerant leak tested to determine conformance with 3.6. The test for leakage shall be made with a halide leak detector or electronic leak detector. Any refrigerant leak shall constitute failure of this test.

4.4.4 Refrigerant moisture test. The liquid moisture and sight indicator shall be checked for dryness in conformance with 3.6 as shown by the component manufacturer's color chart data. A reading indicating moisture shall constitute failure of this test.

4.4.5 Electromagnetic compatibility. When electromagnetic compatibility is required, the production unit shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish the contracting office written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and requirements meet MIL-STD-461.

5. PACKAGING

5.1 Levels A, B, and C. Each refrigerator or food warmer cabinet or combination thereof shall be preserved, packaged, packed and marked for each level of protection in accordance with MIL-R-12323 (see 6.2).

6. NOTES

6.1 Intended use. The refrigerator sections covered by this specification are intended for holding food at 38°F (3.3°C) prior to serving. The food warmer sections are intended to hold food at warming temperatures from 150°F (65.5°C) to 160°F (71.1°C) prior to serving.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, style, inclosure, duty, or model cabinet required (see 1.2).
(c) When mobile racks are not required (see 3.4).
(d) Location in outer shell for refrigerant pipe and electrical connections when other than manufacturer's standard (see 3.5).
(e) Electrical characteristics, if other than specified (see 3.5.1).
(f) Whether condensate evaporator is required (see 3.5.4).
(g) When recessed door handles are required (see 3.5.5).
(h) When self-closing doors are required (see 3.5.5).
(i) When bumpers are required on model A or model B cabinets (see 3.5.8).
(j) When open-type compressor is required (see 3.6).
(k) When electromagnetic compatibility is required (see 3.6.4).
(l) Selection of applicable levels of preservation and packaging and packing (see 5.1).

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Custodian:

Army - GL

Review activity:

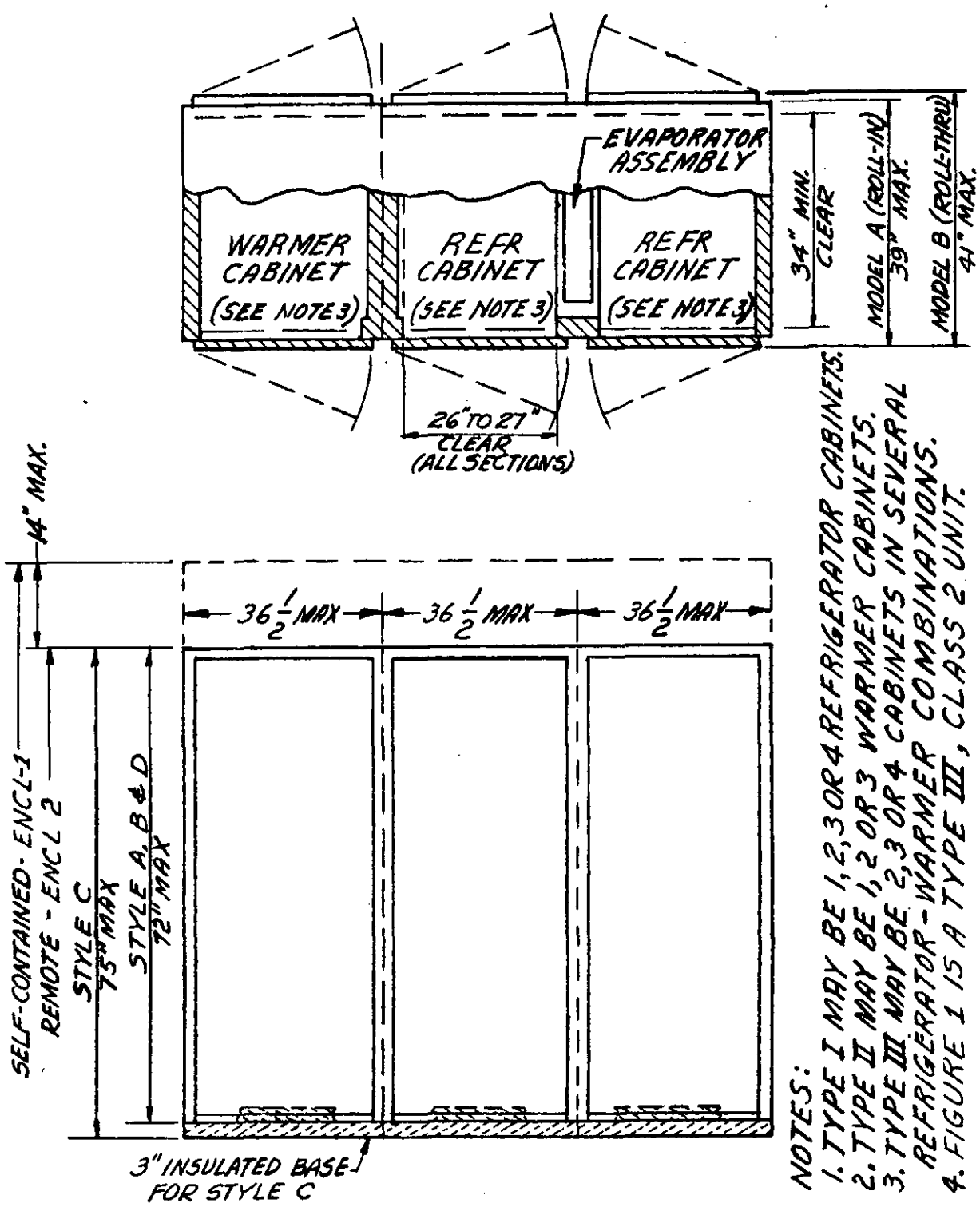
Army - MD

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NOTES:
1. TYPE I MAY BE 1, 2, 3 OR 4 REFRIGERATOR CABINETS.
2. TYPE II MAY BE 1, 2 OR 3 WARMER CABINETS.
3. TYPE III MAY BE 2, 3 OR 4 CABINETS IN SEVERAL REFRIGERATOR-WARMER COMBINATIONS.
4. FIGURE 1 IS A TYPE III, CLASS 2 UNIT.

FIGURE 1 CABINETS, FOOD SERVICE, WARMER-REFRIGERATOR, ROLL-THRU, ROLL-IN

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-C-43427B(GL) - Cabinets, Food Service, Warmer-Refrigerator, Roll-Thru, Roll-In

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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